Disentangling Ambiguity and Asymmetry in the Attraction Effect: When Comparative Processing Drives Context Dependence

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Abstract

The attraction effect—an increased preference for a target when an inferior decoy is added—has been widely replicated yet remains strikingly fragile across seemingly minor changes in stimulus construction and task demands. A common intuition is that difficult inter-attribute trade-offs make people lean more on dominance, but this conflates two distinct ingredients: ambiguity between the core options (Target vs. Competitor) and asymmetric ease of decoy-related comparisons. Across three experiments, we disentangle these components and show that attraction arises when—and only when—decoy-related comparisons are selectively easier once core-option ambiguity is in place. Experiment 1 establishes robust attraction in a mixed perceptual–numerical choice task (circle area + numerical price), confirming that appropriately constructed perceptual stimuli produce positive asymmetric dominance. Experiment 2 uses an adjustment paradigm to measure inter-attribute "equivalence regions" and demonstrates that the same trade-off difficulty that supports attraction also induces broad competitor-referenced regions that often encompass both target and decoy, especially for circle–price stimuli relative to the bar stimuli. Experiment 3 then holds Target–Competitor difficulty approximately constant while manipulating the relative ease of decoy comparisons with controlled fraction triplets, revealing that attraction and its response-time signature emerge precisely when decoy-related comparative asymmetry is high. These results locate the source of the attraction effect in asymmetric comparative processing rather than structural dominance alone, reconciling its robustness in principle with its empirical elusiveness in practice.

Keywords: Preference reversals, attraction effect, asymmetric dominance

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General Introduction

The attraction effect (also called the asymmetric dominance effect) refers to an increase in the choice share of a target option when an inferior decoy is added to a binary choice set. This context effect is a canonical violation of context-independent choice and has been demonstrated across domains and task classes (Huber et al., 1982; Trueblood et al., 2013).

Yet the attraction effect is also empirically fragile: modest changes in stimulus construction, presentation format, task demands, or time pressure can attenuate or eliminate the effect, contributing to the broader “elusiveness” of context effects (Frederick et al., 2014a; Spektor et al., 2021). This fragility is especially salient in perceptual stimuli, where different stimulus families (e.g., bars/rectangles vs. star-like) often yield different outcomes under superficially similar dominance designs (Rath et al., 2025).

A prominent explanation for this variability appeals to inter-attribute trade-off difficulty: when attributes resist commensuration or are difficult to integrate, decision makers may rely more on dominance checks, and decoys may be easier to reject than core alternatives (Hedgcock & Rao, 2009; Luce et al., 2001). However, the key mechanistic question is not whether trade-offs are difficult “in general,” but what makes dominance functionally asymmetric—i.e., why the decoy ends up favoring the target rather than simply adding noise or ambiguity across comparisons.

A concrete example of this identification problem comes from commensurate-stimulus paradigms (e.g., bars) embedded in richer factorial designs, where null or “reversed” effects have been reported under some display/analysis choices. Re-analyses that jointly model multiple design factors suggest that the decoy-designated “target” signal can be best characterized as null once other manipulations (including those that affect baseline discriminability between the core options) are accounted for (Supplementary Material; Rath et al. (2025)). This pattern is consistent with the possibility that commensurate constructions often minimize the prerequisite context in which a decoy can exert leverage, without necessarily diagnosing whether decoy-related asymmetry is causally absent. In that sense, much existing evidence is informative about whether the prerequisite is present, while leaving the causal role of decoy-related comparative asymmetry underdetermined.

We propose that much of the literature inadvertently conflates two distinct ingredients that can both be modulated by trade-off difficulty. The first ingredient is ambiguity/indifference between the core options (Target vs. Competitor): the decision maker lacks a strong prior preference that would render the decoy largely irrelevant (Huber et al., 2014). The second ingredient is comparative asymmetry in decoy processing: the Target–Decoy comparison is selectively easier or more decisive than the Competitor–Decoy comparison (Huber et al., 1982; Rath et al., 2025), allowing dominance to be exploited as a “path of least resistance.”

This yields a logic-gate framing in which attraction depends on two inputs: A (core-option ambiguity) and S (decoy-related comparative asymmetry). If attraction requires both inputs—an AND gate—then attraction should emerge primarily when $(A, S) = (1, 1)$, and should be absent when either ingredient is missing. If, instead, attraction is driven by only one ingredient (an OR-like account), then observing attraction in $(1, 1)$ and its absence in $(0, 0)$ is not diagnostic because both ingredients tend to co-vary under many common manipulations.

The identification problem is that many classic stimulus manipulations effectively toggle A and S together, producing mostly the “diagonal” cases. Commensurate/easy constructions often yield $(0, 0) \Rightarrow 0$: the Target–Competitor trade-off is readily resolvable (low ambiguity) and decoy-related comparisons do not become selectively decisive (low comparative asymmetry), producing a null attraction effect. Incommensurate/difficult constructions often yield $(1, 1) \Rightarrow 1$: the Target–Competitor trade-off becomes ambiguous while the stimulus/task configuration makes the Target–Decoy relation comparatively more decisive than the Competitor–Decoy relation, producing positive attraction. Because these manipulations bundle A and S, the observed mapping $(0, 0) \Rightarrow 0$ and $(1, 1) \Rightarrow 1$ can be explained by multiple causal stories, including ones that misattribute the effect to “difficulty” alone.

To distinguish accounts, we must test the off-diagonal cases. The case $(1, 0)$ (ambiguity present, comparative asymmetry absent) is especially diagnostic for whether comparative asymmetry is the active ingredient once ambiguity is in place. The case $(0, 1)$ is practically difficult to test in a meaningful way because when the Target–Competitor trade-off is easy and preferences are already strong, a decoy typically has little leverage to move choice shares even if a dominance relation is easy to process.

The present manuscript implements this logic across three experiments. Experiment 1 establishes that stimulus constructions associated with difficult inter-attribute trade-offs can reliably produce positive attraction in a mixed perceptual–numerical paradigm, consistent with the $(1, 1) \Rightarrow 1$ corner that dominates much of the prior literature. Experiment 2 then shows why this evidence is not mechanistically diagnostic: using an adjustment paradigm, we measure trade-off geometry directly and find that trade-off difficulty can induce broad regions of subjective equivalence that often encompass multiple options simultaneously, thereby confounding ambiguity in the core trade-off with ambiguity in decoy-related comparisons. Experiment 3 removes this confound by holding Target–Competitor comparison difficulty approximately constant while manipulating decoy-related comparative asymmetry (via controlled fraction stimuli), enabling a within-subject test of $(1, 1)$ versus $(1, 0)$.

Across these studies, our central claim is that ambiguity/indifference is the prerequisite context, but asymmetric comparative processing is the causal lever that produces the attraction shift when that context is in place. This process-level account aligns with recent perceptual results (Rath et al., 2025) showing that apparent inconsistencies track whether stimuli instantiate genuine, item-wise dominance asymmetry and whether dominance relations are realized in comparison, rather than being merely definable on paper.

Experiment 1: Perceptual Attraction Effects

Introduction

Given the mixed empirical record of attraction effects and their sensitivity to stimulus construction, we first sought to establish whether asymmetric dominance reliably produces attraction effects in perceptual decision-making across a range of stimulus configurations. Although recent work has demonstrated robust perceptual attraction effects when dominance relations are appropriately specified (Rath et al., 2025), it remains important to characterize the scope of these effects and the stimulus classes in which they arise.

Experiment 1 therefore served a descriptive purpose. We used a mixed perceptual–numerical decision task in which each option combined a perceptual attribute (visual area) with a numerical attribute (price), and participants were instructed to choose the option with the lowest price per unit area. Following standard asymmetric-dominance designs, we introduced a third option constructed to be asymmetrically dominated by the target but not by the competitor, and we included the canonical decoy types (range, frequency, and range–frequency) (Huber et al., 1982).

This experiment was not designed to adjudicate between competing explanations of the attraction effect, nor to isolate the source of dominance asymmetry. Instead, it established the empirical phenomenon in the present paradigm and delineated the conditions under which asymmetric dominance influences choice. These results provided the foundation for Experiment 2, which directly measured the structure of inter-attribute trade-offs underlying such decisions, and Experiment 3, which isolated decoy-related comparative asymmetry under controlled conditions. In terms of the framework introduced above, Experiment 1 establishes the phenomenon but does not yet distinguish whether attraction reflects ambiguity between the core options or asymmetries in decoy-related comparisons.

Methods

Participants

Thirty-six participants (mean age = 21.11 years, SD = 3.42; 27 male, 9 female) with normal or corrected-to-normal vision took part in the experiment after providing informed consent. Six participants were excluded for failing catch trials (accuracy < 0.8), leaving a final sample of 30 participants for analysis. All procedures were approved by the relevant institutional ethics committee.

Stimuli

Each trial displayed three black-filled circular discs on a white background, arranged in a triangular configuration around the center of the screen with vertical positions jittered across trials. Figure 2 illustrates an example trial. Each disc represented a hypothetical alloy, with its price displayed as a numerical value at the center of the disc in arbitrary units.

The decision-relevant attributes were the perceptual area of the disc and its numerical price. Participants were instructed to choose the disc with the lowest price per unit area, requiring integration of a perceptual attribute (area) with a symbolic numerical attribute (price).

Stimuli were generated following procedures adapted from Trueblood et al. (2013). One set of discs was sampled from a bivariate normal distribution with a mean area of 28,000 pixels and a mean price of 28 units. The variances for area and price were 28,000 pixels and 28 units, respectively, with no correlation between dimensions. A second set of discs was matched in price per unit area but had twice the area of the first set. These two sets served as the target and competitor, counterbalanced across trials.

Decoy discs were generated such that, in the attribute space, they were positioned closer to one of the two core options, creating two contextual conditions. All three standard decoy types—range, frequency, and range-frequency decoys—were included (Huber et al., 1982). Across conditions, decoys were always asymmetrically dominated by the target but not by the competitor.

Design and Procedure

Participants completed 36 trials in each contextual condition, resulting in 72 experimental trials. In addition, 12 catch trials were included, in which one option clearly had the lowest price per unit area. The full set of 84 trials was presented in randomized order.

On each trial, participants selected one of the three discs using the left, up, or right arrow keys. They were instructed to respond as quickly and accurately as possible. No time limits were imposed.

Measures

Choice frequencies for the target, competitor, and decoy were recorded on each trial. Context effects were quantified using the equal-weights version of the Relative Choice Share of the Target (RST_{EW}) (Katsimpokis et al., 2022). Response times were recorded for exploratory analyses.

Results

Context effects were quantified using the equal-weights version of the Relative Choice Share of the Target (RST_{EW}; (Katsimpokis et al., 2022)). RST_{EW} captures the relative frequency with which the target is chosen over the competitor across decoy-favoring contexts, with a value of 0.5 indicating no context effect. RST_{EW} was computed as:

$$RST_{\text{EW}} = \frac{1}{2}\left(\frac{T_X}{T_X + C_X} + \frac{T_Y}{T_Y + C_Y}\right),$$

where T_X and C_X denote the number of target and competitor selections, respectively, when the decoy favored option X, and T_Y and C_Y denote the corresponding counts when the decoy favored option Y.

Across participants, the mean RST_{EW} was $M = 0.591$ ($SD = 0.109$), exceeding the null value of 0.5. A one-sample one-tailed t-test indicated that this difference was statistically significant, $t(29) = 4.593$, $p < .001$, with a large effect size (Cohen's $d = 0.839$).

A Bayesian one-sample t-test further supported this result, yielding strong evidence in favor of the alternative hypothesis that $\text{RST}_{\text{EW}} > 0.5$ (Bayes factor $\text{BF}_{10} = 273.75$, Cauchy prior with $r = 0.4$). Figure **??** displays the distribution of RST_{EW} values across participants.

Response time analyses were conducted for exploratory purposes. Mean response times did not differ systematically across decoy contexts and are therefore not discussed further here.

Discussion

Experiment 1 demonstrated a robust attraction effect in a mixed-attribute decision task involving both perceptual and numerical information. Participants reliably preferred the target over the competitor when an asymmetrically dominated decoy was introduced, even though the decision required integrating a perceptual attribute (area) with a symbolic numerical attribute (price) to compute price per unit area. This finding replicates prior demonstrations of attraction effects in non-preference-based perceptual tasks (Rath et al., 2025; Trueblood et al., 2013) and extends them to settings that combine perceptual and numerical attributes.

Importantly, Experiment 1 was not designed to isolate the source of dominance asymmetry. As in many perceptual and criterion-based attraction studies, the target and competitor were matched on the decision-relevant criterion, ensuring no objective advantage prior to the introduction of the decoy. However, this design choice also leaves open whether the observed attraction effect reflects (i) weak or unstable baseline preference between the target and competitor, (ii) the selective realization of dominance in comparisons involving the decoy, or some combination of the two. In the terms developed in the General Introduction, Experiment 1 establishes attraction in the present paradigm but does not yet separate core-option ambiguity from decoy-related comparative asymmetry.

Thus, while Experiment 1 establishes that attraction effects can arise in mixed perceptual–numerical decision spaces, it does not adjudicate between alternative explanations of how dominance exerts its influence on choice. In particular, the integration of perceptual and numerical attributes may introduce trade-off uncertainty that affects multiple pairwise comparisons simultaneously, making it unclear whether the observed attraction reflects a selective asymmetry in decoy-related comparisons or a broader “region” of indifference that also encompasses the target–competitor trade-off. Experiment 2 therefore directly examines the geometry of inter-attribute trade-offs underlying such decisions, providing a diagnostic test of whether trade-off uncertainty is selective in the manner required to support asymmetric dominance, or instead broad in a way that confounds multiple comparisons.

Experiment 2: Diagnosing Inter-Attribute Trade-Off Geometry

Introduction

Experiment 1 established that our mixed perceptual–numerical stimulus family (circle area + price) produces a reliable attraction effect, but that finding alone does not identify why the decoy shifts choice toward the target. A common intuition is that “trade-off difficulty” (e.g., integrating a perceptual quantity with a symbolic number) makes people lean more on dominance checks, but asymmetric dominance requires more than global difficulty: it requires selectivity, such that the Target–Decoy comparison is systematically more decisive than the Competitor–Decoy comparison. If the same manipulation that increases trade-off difficulty also creates broad indifference between the core options (Target vs. Competitor) and simultaneously blurs decoy-related comparisons, then observing attraction becomes mechanistically ambiguous rather than explanatory.

Experiment 2 addresses this identification problem by measuring the geometry of inter-attribute trade-offs directly, rather than inferring difficulty from choice shares. We use an adjustment (matching) paradigm in which participants construct subjective equivalence between two options by adjusting one attribute until the pair appears equally good under the same task criterion used in the choice task. This design lets us estimate the dispersion (width) and stability of equivalence judgments—i.e., how “tight” versus “mushy” participants’ equivalence regions are—within each stimulus family.

Critically, Experiment 2 asks whether trade-off uncertainty is selective in the way needed to support asymmetric dominance, or whether it is broad enough that multiple distinct options can fall into the same competitor-centered equivalence region. To make this diagnostic concrete, we compare equivalence construction in circle+price stimuli (area traded off against price) to a commensurate perceptual baseline (bars, where the criterion reduces to a sum), and we quantify how often the target and decoy would be “close enough” to the competitor under each stimulus family’s measured equivalence width. These measurements motivate Experiment 3’s more controlled test, where target–competitor difficulty is held roughly constant while decoy-related comparative asymmetry is manipulated directly.

Methods

Participants

Participants were recruited from [institutional participant pool/university community/online platform] and provided informed consent prior to participation. All procedures were approved by the relevant institutional ethics committee.

Data collection followed a sequential Bayesian sampling plan. A minimum of $N = 20$ participants were collected before interim analyses were conducted. After each additional participant, the Bayes factor associated with the primary hypothesis (structural ambiguity difference between stimulus families) was updated. Data collection was terminated when either (a) $\mathrm{BF}_{10} \geq 10$ (evidence for the alternative), (b) $\mathrm{BF}_{01} \geq 10$

(evidence for the null), or (c) the maximum sample size of $N_{\max} = 70$ was reached.

The maximum sample size was determined via conservative power analysis (see Sample Size Determination below). All stopping criteria were specified a priori.

Participants were excluded prior to analysis if they failed more than two catch trials per stimulus block (relative error greater than 20%), did not complete the session, or exhibited clear non-engagement with the adjustment task (e.g., invariant responses across trials). All exclusion criteria were determined a priori.

Sample Size Determination

Sample size planning was based on the primary hypothesis concerning structural ambiguity differences between stimulus families. A conservative frequentist power analysis for the exact McNemar test (two-tailed, $\alpha = .05$, power = .90) assumed that 60% of participants would exhibit circle-only structural ambiguity and 20% would exhibit bar-only structural ambiguity (odds ratio = 3). Under these assumptions, the required fixed sample size was $N = 70$.

Because confirmatory inference was conducted using a sequential Bayesian framework, this value was treated as an upper bound ($N_{\max} = 70$). A Bayesian factor design analysis (simulation-based) was conducted during planning to examine expected stopping behavior under these assumptions.

Design and Procedure

The experiment employed a fully within-subjects design. Each participant completed 80 bar trials and 80 circle trials, along with 20 catch trials (10 per stimulus family). Block order (bar first vs. circle first) was counterbalanced with probability 0.5 across participants. Trials were randomized within blocks using a Fisher–Yates shuffle with a time-stamped random seed to ensure reproducibility. A brief break separated the two stimulus blocks.

On each trial, the reference option was displayed on the left and the adjustable option on the right. The relevant decision rule (sum for bars; ratio for circles) was displayed at the top of the screen. Participants adjusted the slider until satisfied that both options were equivalent in overall value and confirmed their response. No time limit was imposed. Reaction time was recorded from trial onset to confirmation using high-resolution timing.

The primary confirmatory hypothesis concerned differences in structural ambiguity between stimulus families. Sequential Bayesian evaluation of this hypothesis was conducted only after full completion of each participant's data. No interim analyses were conducted within participants. Block order and trial order randomization were independent of stopping decisions.

Operational Definition of Inter-Attribute Trade-Off Geometry

Inter-attribute trade-off geometry was operationalized using an adjustment paradigm. On each trial, participants adjusted one attribute of an option until it was judged equivalent in overall value to a fixed reference option.

For each participant and stimulus family, equivalence dispersion was quantified as the standard deviation of relative criterion error (VE_{rel}). An equivalence region was defined as:

$$\text{Equivalence Band} = k \times VE_{rel},$$

where $k = 1.5$.

An offset was classified as included if its magnitude did not exceed this band. The multiplier $k = 1.5$ was selected a priori as a moderate tolerance threshold. Sensitivity analyses conducted on pilot data demonstrated that the primary structural effect remained positive across $k \in [1.0, 2.0]$, supporting robustness of this choice.

Measures

For each trial, the final adjusted value, true match value, absolute error (AE), relative error, and reaction time were recorded. Absolute error was defined as

$$\mathrm{AE} = |\text{response} - \text{true match}|.$$

Relative error scaled this deviation by the true match value. Additional quality-control metrics included boundary-hit frequency and invariant responding across trials.

For structural classification, predefined target (7%) and decoy (10%) offsets were evaluated against each participant's equivalence band. Participants were classified as exhibiting joint structural ambiguity when both offsets were included within their equivalence region.

Results

All analyses followed the preregistered plan. The primary confirmatory hypothesis (H2) concerned structural ambiguity differences between stimulus families. The dispersion hypothesis (H1) was secondary. Bayesian and frequentist results are reported together for transparency.

A total of 15 participants completed the study. Two participants were excluded based on the preregistered catch-trial criterion, yielding a final analysis sample of $N = 13$.

H1: Equivalence Dispersion

Equivalence dispersion (VE_{rel}) was computed separately for bar and circle stimuli for each participant. As preregistered, we tested whether dispersion was larger for circle stimuli.

Circle stimuli exhibited greater dispersion than bar stimuli. A paired-samples t-test confirmed this difference, $t(12) = 3.543$, $p = .0040$, Cohen's $d_z = 0.983$, indicating a large effect.

The corresponding Bayesian paired-samples t-test (Cauchy prior, $r = 0.707$) yielded $\mathrm{BF}_{10} = 10.188$, indicating strong evidence that dispersion was greater for circle stimuli than for bar stimuli. The posterior mean difference was 0.405.

These results indicate that circle stimuli substantially increase inter-attribute trade-off dispersion relative to bar stimuli.

H2: Structural Ambiguity

For each participant and stimulus family, predefined target (7%) and decoy (10%) offsets were evaluated against the participant's equivalence band ($k = 1.5$). Participants were classified as exhibiting joint structural ambiguity when both offsets fell within their equivalence region.

Circle stimuli produced joint structural ambiguity in 100% of participants, whereas bar stimuli produced joint structural ambiguity in 15.4% of participants. The risk difference was 0.846.

A McNemar test comparing paired classifications indicated $\chi^2(1) = 9.091$, $p = .0026$.

The preregistered Bayesian paired binary analysis yielded $\mathrm{BF}_{10} = 170.667$, indicating very strong evidence that circle stimuli produce greater structural ambiguity than bar stimuli. The discordant pattern was $k = 11$ circle-only classifications and $m = 0$ bar-only classifications.

Continuous Distance Analysis (Robustness Check)

To complement the binary classification, we examined continuous standardized distances of the target and decoy offsets from each participant's equivalence boundary. These analyses yielded qualitatively similar conclusions, indicating that circle stimuli reduced the standardized separation between offsets and equivalence boundaries relative to bar stimuli (see Supplementary Section S2).

Sensitivity to Equivalence Multiplier

Sensitivity analyses varying the equivalence multiplier ($k \in [1.0, 2.0]$) demonstrated that the structural effect remained positive across the full tested range. Across this interval, the proportion difference ranged from 0.50 to 0.92, and Bayes factors ranged from 9.14 to 170.67 (see Supplementary Figure S3). These results indicate that the primary structural finding does not depend on the specific choice of $k = 1.5$.

Sequential Evidence Accumulation

Sequential Bayes factor trajectories for the primary hypothesis are shown in Supplementary Figure S4. The evidence for H2 exceeded $\mathrm{BF}_{10} = 100$ within the observed sample size.

Experiment 3: Fraction Choice and Comparative Processing Asymmetry

Introduction

Experiments 1 and 2 established two critical points. First, attraction effects can arise robustly using incommensurate attributes such as perceptual and numerical attributes. Second, manipulations intended to induce asymmetry through inter-attribute trade-off difficulty can simultaneously affect multiple pairwise comparisons. In particular, when options fall within regions of subjective equivalence, increasing difficulty for decoy–competitor comparisons often also increases difficulty for target–competitor comparisons. This coupling introduces a confound that obscures the source of attraction: observed effects may arise either from asymmetric decoy-related comparisons or from changes in the difficulty of comparing the core options.

Experiment 3 was designed to resolve this confound by isolating asymmetric comparative processing involving the decoy while holding constant the difficulty of target-competitor comparison. To achieve this, we moved to a numerical domain using fraction stimuli, which afford precise control over pairwise relationships among options. Fractions allow comparative distances between option pairs to be manipulated independently, enabling selective variation in the ease of target–decoy and competitor–decoy comparisons without altering the relative comparability of the target and competitor.

This design permits a direct test of the central hypothesis of the present work: that attraction arises when—and only when—comparisons involving the decoy are processed asymmetrically. By controlling target–competitor comparison difficulty and systematically manipulating decoy-related asymmetry, Experiment 3 assesses whether asymmetric dominance exerts its influence through selective decoy elimination rather than through global trade-off difficulty or extended deliberation. In addition, trial-level response-time analyses allow us to dissociate the role of deliberation in decoy elimination from its role in target–competitor preference formation.

Method

Participants

Fifty-five participants (age range XX–XX years; XX female) were recruited from an online participant pool. All participants provided informed consent prior to participation and were compensated according to institutional guidelines. The study was approved by the institutional ethics committee. Data from all participants were included in the analyses reported below.

Stimuli

Each trial consisted of a choice among three options: a *Target* (T), a *Competitor* (C), and a *Decoy* (D). Each option was defined by two numerical attributes, represented as fractions. Fractions were chosen to allow precise control over pairwise relationships among options while maintaining a common metric across attributes.

The decoy was constructed to be asymmetrically dominated by the target but not by the competitor. Critically, stimulus construction allowed selective manipulation of the relative ease of Target–Decoy and Competitor–Decoy comparisons while holding Target–Competitor comparison difficulty approximately constant. This manipulation formed the basis of the $\triangle$Diff condition described below.

Figure 5 shows RST as a function of response-time quartile separately for each $\triangle$Diff and block-order condition. Although RST sometimes appeared to increase with deliberation time, this trend was inconsistent across conditions, motivating a decomposition of choice into decoy elimination and target–competitor preference.

Operational Definition of Comparative Ease

To operationalize the relative ease of pairwise comparisons among options, we quantified comparative discriminability using a normalized geometric-mean distance measure across attributes. For any two options i and j, each defined by attributes a_1 and a_2, comparative ease was defined as:

$$\text{Ease}(i,j) = \sqrt{\left|\log\left(\frac{a_{1i}}{a_{1j}}\right)\right| \cdot \left|\log\left(\frac{a_{2i}}{a_{2j}}\right)\right|}$$

This measure captures joint proportional separation between two options across both attributes while remaining invariant to scale transformations. Larger values correspond to greater separability and, consequently, easier pairwise comparison.

For each trial, three pairwise ease values were computed: Target–Decoy (TD), Competitor–Decoy (CD), and Target–Competitor (TC). The ΔDiff manipulation was defined as the relative difference between TD and CD ease:

$$\Delta\text{Diff} = \text{Ease(TD)} - \text{Ease(CD)}$$

In the High ΔDiff condition, TD comparisons were substantially easier than CD comparisons. In the Low ΔDiff condition, this asymmetry was reduced, rendering TD and CD comparisons more similar in ease. Across both conditions, TC ease was held approximately constant by construction.

The geometric-mean formulation and associated constraints were used solely to guide stimulus selection and ensure controlled manipulation of pairwise comparative asymmetry. These heuristics were used solely to constrain stimulus selection and do not constitute claims about the strategies participants employed during the task. Full details of the heuristic criteria, weighting schemes, sampling procedure, and threshold values are provided in the Supplementary Materials.

Design and Procedure

The experiment employed a within-subjects design with two primary conditions corresponding to levels of comparative asymmetry: High and Low ΔDiff. Trials were presented in two blocks, one per condition. Block order was counterbalanced across participants (High–Low vs. Low–High). Each block contained an equal number of trials, and the same Target–Competitor pairs were used across conditions, differing only in the construction of the decoy.

On each trial, the three options were displayed simultaneously on the screen in randomized spatial order. Participants were instructed to select the option they preferred by clicking on it. No time limit was imposed. Choice and response time (RT) were recorded on every trial.

Response Time Processing

Response times were measured from stimulus onset to choice selection. Trials with missing responses or non-positive RTs were excluded prior to analysis (less than X% of trials). RTs were log-transformed to reduce skew. For descriptive analyses, trials were additionally grouped into participant-wise RT quartiles.

Analytic Strategy

Analyses proceeded in three stages. First, overall choice proportions and the relative share of the target (RST) were computed to assess attraction effects across conditions. Second, decoy choice was modeled at the trial level using hierarchical logistic regression with log-transformed RT, ΔDiff condition, and block order as predictors, including participant-specific random intercepts and slopes. Third, analyses of target choice were restricted to trials in which participants selected either the target or the competitor, allowing assessment of target–competitor preference conditional on decoy elimination.

All hierarchical models were estimated in a Bayesian framework using weakly informative priors. Posterior inference focused on credible intervals and region-of-practical-equivalence (ROPE) analyses to assess evidence for practically meaningful effects.

Results

Overall Choice Patterns

Across conditions, the fraction task reproduced canonical attraction patterns when comparative asymmetry was high. The relative share of the target exceeded chance levels under high △Diff, whereas under low △Diff, target and competitor shares were approximately balanced, yielding RST values near 0.5. This pattern was observed across block orders, indicating that the effect was not driven by presentation sequence.

Decoy Elimination and Response Time

To examine the role of deliberation in decoy processing, we modeled decoy choice at the trial level using a hierarchical logistic regression with RT as a predictor. Across all conditions, the probability of choosing the decoy decreased sharply with increasing response time. This effect was robust across △Diff and block order and showed substantial consistency across participants.

Figure 6 illustrates the conditional effect of RT on decoy choice probability. Faster responses were associated with frequent decoy selection, whereas slower responses were overwhelmingly dominated by choices between the target and competitor. This finding indicates that decoy elimination is a time-sensitive process that unfolds during deliberation.

Target–Competitor Preference Conditional on Decoy Elimination

The critical test of the attraction mechanism concerns relative preference between the target and competitor once the decoy has been eliminated. To assess this, we restricted analyses to trials in which participants chose either the target or the competitor and modeled target choice using a hierarchical logistic regression with RT, △Diff, block order, and their interactions as predictors.

Across all conditions, posterior estimates of RT slopes were small, and 95% credible intervals consistently spanned zero. Figure 7 summarizes posterior mean RT effects and credible intervals for each ΔDiff and block-order condition. No condition showed reliable evidence for a systematic increase or decrease in target preference with deliberation time.

Region-of-practical-equivalence (ROPE) analyses further supported this conclusion. In several conditions, the majority of posterior mass for the RT slope lay within a narrow ROPE around zero, providing evidence for the absence of practically meaningful RT effects. In remaining conditions, posterior mass was broadly distributed but did not support reliable deviations from zero.

Decomposing Apparent RT Effects

Descriptive analyses of response-time quartiles revealed that target choice frequency often increased with RT, while decoy choice frequency decreased sharply. Figure 8 shows the distribution of target, competitor, and decoy choices across RT quartiles. Importantly, increases in target frequency largely mirrored decreases in decoy frequency, rather than shifts in target–competitor preference.

This decomposition clarifies why RT-based analyses sometimes suggest that attraction strengthens with deliberation. Apparent increases in target choice with RT primarily reflect progressive decoy elimination, not amplification of target preference relative to the competitor.

Summary of Findings

Experiment 3 yields three key results. First, attraction effects emerge reliably when comparative processing is asymmetric, as indexed by ΔDiff, and are attenuated when such asymmetry is reduced. Second, deliberation time strongly predicts decoy elimination across all conditions, consistent with time-sensitive dominance detection. Third, conditional on decoy elimination, target–competitor preference shows little evidence of systematic modulation by deliberation time or comparative asymmetry.

Together, these findings suggest that attraction in fraction choice arises from asymmetric comparative processing that governs decoy elimination, rather than from gradual strengthening of target preference during extended deliberation. This account provides a mechanistic reinterpretation of prior findings that link time pressure and response speed to the attenuation of attraction effects (Spektor et al., 2021).

Discussion

Experiment 3 demonstrates that attraction in fraction choice arises from asymmetric comparative processing that governs decoy elimination rather than from extended deliberation between the target and competitor. Although response time strongly predicted whether the decoy was chosen, it did not reliably modulate relative preference between the target and competitor once the decoy was excluded. Apparent increases in target choice with longer response times were largely attributable to progressive decoy elimination, as reflected in parallel increases in both target and competitor choice frequencies. These findings indicate that response time influences which options remain under consideration, but does not systematically bias preference among the remaining options. Accordingly, attraction in this task reflects asymmetries in how decoy-related comparisons are resolved, rather than gradual amplification of target preference during deliberation.

General Discussion

The attraction effect has long occupied a paradoxical position in decision research. On the one hand, the addition of an asymmetrically dominated decoy reliably shifts preferences toward its dominating target across domains and species (Huber et al., 1982; Trueblood et al., 2013). On the other hand, the effect often proves fragile, disappearing under modest changes in task structure, stimulus representation, or temporal constraints (Frederick et al., 2014b; Spektor et al., 2021). The present work resolves this tension by identifying the source of asymmetry in the asymmetric dominance effect: not in dominance per se, nor in global trade-off difficulty, but in asymmetric comparative processing involving the decoy.

Across three experiments, we show that attraction depends critically on whether decoy-related comparisons are selectively easier than competing comparisons. When such asymmetry is present, attraction emerges robustly; when it is absent or confounded, attraction becomes unstable or disappears. This account reconciles the apparent robustness of the effect with its empirical elusiveness and provides a principled reinterpretation of prior findings.

Asymmetry Is Comparative, Not Structural

Classical accounts often treat dominance asymmetry as a structural property of the option set: the decoy is worse than the target on all attributes but not worse than the competitor (Huber et al., 1982; Simonson, 1989). Our results show that this structural definition is insufficient. What matters is whether dominance asymmetry is *comparatively realized*—that is, whether the cognitive system can easily detect and exploit the dominance relation during pairwise evaluation.

Experiment 1 demonstrated that robust attraction effects arise in perceptual choice when stimulus construction ensures that decoy–target comparisons are comparatively salient. However, Experiment 2 revealed that commonly invoked explanations based on inter-attribute trade-off difficulty or incommensurability (Chang, 2013; Walasek & Brown, 2023) can confound multiple pairwise relationships. Specifically, when trade-off difficulty induces broad regions of subjective equivalence, both target–competitor and competitor–decoy comparisons become difficult for the same structural reasons. In such cases, trade-off difficulty does not selectively restore dominance asymmetry but instead obscures its comparative realization. I.e., when both target–competitor and competitor–decoy comparisons become difficult, the source of any observed attraction effect becomes ambiguous.

Experiment 3 resolved this confound by holding target–competitor comparison difficulty constant while selectively manipulating the relative ease of target–decoy versus competitor–decoy comparisons. Under these conditions, attraction emerged precisely when decoy-related asymmetry was introduced. This demonstrates that dominance influences choice not merely by its formal definition, but by how selectively it guides comparative processing.

Decoy Elimination and Preference Formation

A second contribution of the present work is the dissociation of decoy elimination from target–competitor preference formation. Across conditions, response time robustly predicted a reduction in decoy choice, consistent with the view that extended processing facilitates the rejection of dominated alternatives. However, conditional on decoy elimination, response time did not reliably shift preference between the target and competitor.

This finding clarifies a recurring ambiguity in the attraction literature. Increases in target choice are often interpreted as evidence of strengthened preference for the target. Our analyses show that such increases can arise purely from the removal of the decoy from consideration, leaving the relative balance between target and competitor unchanged. This decomposition aligns with critiques emphasizing the importance of analyzing full choice distributions rather than aggregated target shares (Katsimpokis et al., 2022; Liew et al., 2016). Further, all reported effects were robust to out-of-sample validation: hierarchical models exhibited stable predictive performance under Pareto-smoothed leave-one-out cross-validation, with no evidence that inferences were driven by influential trials.

Reinterpreting Time Pressure Effects

The present findings also shed new light on the effects of time pressure. Prior work has consistently shown that both external and internal time pressure weaken or eliminate attraction effects (**marini2019**; Spektor et al., 2021). This pattern has often been interpreted as evidence that attraction requires slow, deliberative, compensatory processing.

Our results suggest a more specific interpretation. Time pressure disrupts attraction not because it eliminates comparison altogether, but because it truncates the selective comparative processing that allows decoy-related asymmetries to influence choice. Under limited time, decoy elimination is incomplete, and dominance relations fail to exert their asymmetric influence. This account is consistent with dynamic models of multialternative choice, which predict that context effects emerge only after sufficient comparative evidence has accumulated (Evans et al., 2019; Roe et al., 2001; Usher & McClelland, 2004).

Importantly, this interpretation explains why cognitive load manipulations often fail to eliminate attraction effects (**WedellHayes2022**; Wedell, 1991): generic effort is not the critical factor. Rather, what matters is whether the task affords sufficient time for selective pairwise comparisons to unfold.

Implications for Theories of Context Effects

By locating the source of attraction in asymmetric comparative processing, the present work bridges competing theoretical traditions. It complements sequential sampling and dynamic accumulation models that emphasize comparison dynamics (Bhatia, 2013; Roe et al., 2001; Trueblood et al., 2014), while also aligning with reason-based and comparison-based accounts of choice (Russo & Dosher, 1983; Simonson, 1989). At the same time, it cautions against explanations that appeal exclusively to attribute trade-off difficulty or incommensurability, which may conflate multiple sources of comparison difficulty.

More broadly, this perspective helps explain why attraction effects appear across diverse species and tasks (Bateson et al., 2003; Parrish et al., 2015), yet remain sensitive to seemingly minor design details. Dominance asymmetry is not a static property of the choice set; it is a process-dependent outcome of how comparisons are structured and realized.

Conclusion

The attraction effect is neither a fragile illusion nor a universal inevitability. It arises when—and only when—comparative processing is asymmetrically structured around the decoy. By disentangling structural dominance from comparative realization, the present work offers a unifying account of why attraction effects are robust in principle yet elusive in practice. This framework provides a foundation for future work on context effects that moves beyond asking whether attraction occurs, toward understanding precisely when and why dominance shapes choice.

Transparency and Data Availability

All data preprocessing steps, analysis code, and materials will be made publicly available on the Open Science Framework upon publication.

Supplementary Methods

S1. Fraction Stimulus Space

All fraction stimuli were drawn from the space of rational numbers with integer numerators and denominators greater than 1 and less than 100, excluding cases in which the numerator equaled the denominator. This space was chosen to avoid trivial equivalences while allowing sufficient flexibility to construct controlled triplets satisfying the comparative constraints described below.

Potential fraction triplets were sampled randomly from this space and evaluated against a set of stimulus-selection criteria. Sampling continued until a fixed set of stimuli meeting all constraints was obtained for each experimental condition.

S2. Operationalization of Holistic Fraction Magnitude

To avoid extreme or degenerate fraction comparisons, we quantified the ease of perceiving the holistic magnitude (or “super-value”) of individual fractions. This quantity was used solely as a stimulus-selection constraint and played no role in the theoretical interpretation of results.

Holistic magnitude was computed as the geometric mean of a set of binary conditions informed by prior work on fraction comparison. To prevent the geometric mean from collapsing to zero when conditions were absent, condition presence was coded as 1.05 and absence as 0.05. The conditions were:

1. The number of common factors between the numerator and denominator was greater than or equal to four.
2. The ratio represented by the fraction lay within ± 0.15 of a whole number.
3. The numerator was perfectly divisible by the denominator.
4. The denominator was divisible by five.

These constraints were used to ensure that all fractions fell within a comparable and interpretable range of magnitudes, thereby reducing unintended variability across trials.

S3. Pairwise Ease-of-Comparison Heuristics

To guide stimulus selection for pairwise comparisons, we defined an ease-of-comparison score for each fraction pair. This score was computed as the geometric mean of multiple comparison-relevant conditions, again using a 0.05 offset to avoid zero values. The conditions were informed by documented regularities in fraction comparison behavior and included:

1. Whether the two fractions shared a common component (numerator or denominator). This factor was weighted four times more heavily than the others.
2. Whether the comparison was gap congruent (i.e., the fraction with the larger numerator–denominator gap was also the correct choice), with the ratio between the gaps exceeding 1.5.
3. Whether the ratio between numerators was less than or equal to 5.05 or lay within ± 0.15 of a whole number.

4. Whether the same criterion held for denominators.
5. Whether the ratio between the holistic magnitudes of the two fractions was either very large (greater than 2) or very small (within ± 0.15 of unity).

These heuristics were used exclusively to rank candidate fraction pairs by comparative separability. They do not imply that participants explicitly relied on these heuristics during the task.

S4. Construction of $\triangle$Diff Conditions

Using the pairwise ease scores, three comparisons were evaluated for each candidate triplet: Target–Decoy (TD), Competitor–Decoy (CD), and Target–Competitor (TC).

In the Low $\triangle$Diff condition, the ratio between TD and CD ease scores was constrained to be less than 2, and a minimum ease threshold of 0.4 was imposed for both comparisons. In the High $\triangle$Diff condition, the ratio between TD and CD ease scores was constrained to be greater than 4, such that the decoy was substantially easier to compare with the target than with the competitor.

Across both conditions, TC ease scores were constrained to lie within a fixed range (ratios between 2 and 5.5) and were further restricted to avoid values within ± 0.15 of a whole-number ratio. These constraints ensured that Target–Competitor comparison difficulty remained approximately constant across $\triangle$Diff conditions.

The logic of this comparative asymmetry manipulation is illustrated schematically in Supplementary Figure 9.

S5. Final Stimulus Set

Following application of all constraints, 30 base fraction triplets were selected. Each triplet was instantiated in two contextual variants (corresponding to the two decoy constructions), yielding 60 trials per $\triangle$Diff condition. The same Target–Competitor pairs appeared in both conditions, differing only in the relative construction of the decoy.

S6. Clarification on Interpretation

All heuristic criteria and geometric-mean–based measures described above were used solely to constrain stimulus selection and implement controlled manipulations of pairwise comparative asymmetry. They do not constitute claims about the strategies participants employed during the task, nor do they imply that participants explicitly computed or accessed these quantities.

S7. Overview and Modeling Rationale

Experiment 3 aimed to isolate the role of asymmetric comparative processing in producing attraction effects while controlling for target–competitor comparison difficulty. To this end, we analyzed choice behavior at the trial level using hierarchical Bayesian models that explicitly distinguished (i) the elimination of the decoy from (ii) preference between the target and the competitor conditional on decoy elimination. This decomposition follows recent recommendations to analyze full choice distributions rather than aggregated target shares and allows separate assessment of mechanisms operating at different stages of decision-making.

All models were estimated in a Bayesian framework with weakly informative priors. Posterior inference relied on credible intervals and region-of-practical-equivalence (ROPE) analyses rather than null-hypothesis significance testing.

S8. Decoy Choice Model

Decoy choice was modeled using a hierarchical logistic regression applied to all trials. Let D_{ij} denote whether participant j selected the decoy on trial i ($D_{ij} = 1$ if the decoy was chosen, 0 otherwise). The model was specified as:

$$D_{ij} \sim \text{Bernoulli}(\pi_{ij}),$$

$$\text{logit}(\pi_{ij}) = \beta_0 + \beta_1 \log(\text{RT}_{ij}) + \beta_2 \Delta\text{Diff}_{ij} + \beta_3 \text{BlockOrder}_j + \beta_4 \log(\text{RT}_{ij}) \times \Delta\text{Diff}_{ij} + u_{0j} + u_{1j} \log(\text{RT}_{ij}),$$

where $\log(\mathrm{RT}_{ij})$ is the log-transformed response time, $\Delta\mathrm{Diff}_{ij}$ denotes the comparison-difficulty manipulation (high vs. low), and $\mathrm{BlockOrder}_j$ indicates block order (HL vs. LH). Participant-specific random intercepts (u_{0j}) and random slopes for logRT (u_{1j}) captured individual differences in baseline decoy avoidance and time sensitivity.

This model assessed whether extended deliberation selectively reduced decoy choice and whether this reduction depended on the asymmetric comparison structure induced by the ΔDiff manipulation.

S9. Target Choice Conditional on Decoy Elimination

To assess whether attraction effects reflect changes in target–competitor preference beyond decoy elimination, target choice was analyzed conditional on trials in which the decoy was not selected. Let T_{ij} denote whether participant j chose the target on trial i, given $D_{ij} = 0$.

$$T_{ij} \mid (D_{ij} = 0) \sim \mathrm{Bernoulli}(\theta_{ij}),$$

$$\mathrm{logit}(\theta_{ij}) = \gamma_0 + \gamma_1 \log(\mathrm{RT}_{ij}) + \gamma_2 \Delta\mathrm{Diff}_{ij} + \gamma_3 \mathrm{BlockOrder}_j + \gamma_4 \log(\mathrm{RT}_{ij}) \times \Delta\mathrm{Diff}_{ij} + v_{0j} + v_{1j} \log(\mathrm{RT}_{ij}).$$

This conditional analysis isolates target–competitor preference formation after the decoy has been eliminated, allowing direct assessment of whether response time or comparison structure shifts relative preference rather than merely suppressing decoy choice.

S10. Prior Specification

All priors were chosen to be weakly informative, providing regularization without favoring either the presence or absence of attraction effects.

Fixed-effect coefficients were assigned normal priors:

$$\beta_k, \gamma_k \sim \mathcal{N}(0, 1),$$

corresponding to modest prior beliefs about plausible log-odds effects.

Standard deviations of participant-level random effects were assigned half-Student-t priors:

$$\sigma \sim \text{Half-Student-}t(3, 0, 1),$$

and correlation matrices were assigned LKJ priors:

$$\mathbf{R} \sim \text{LKJ}(2).$$

These priors regularize extreme individual differences while allowing substantial heterogeneity across participants.

S11. Region of Practical Equivalence (ROPE)

To assess evidence for practically meaningful effects, posterior distributions were evaluated using ROPE analyses. For regression coefficients expressed in log-odds units, the ROPE was defined as $[-0.1, 0.1]$, corresponding approximately to differences in choice probability smaller than 2–3% across the observed range of predictors.

Posterior mass falling largely within the ROPE was interpreted as evidence for a practically negligible effect, whereas posterior mass predominantly outside the ROPE was interpreted as evidence for a meaningful effect. Intermediate cases were classified as inconclusive.

S12. Model Robustness and Predictive Adequacy

As a robustness check, we evaluated the predictive adequacy of all hierarchical models using Pareto-smoothed importance sampling leave-one-out cross-validation (PSIS-LOO; (**Vehtari2017**)). For both the decoy-elimination model and the target-choice model, all Pareto-k diagnostics were below 0.7, indicating that posterior estimates were not driven by influential observations and that the LOO approximation was reliable. The effective number of parameters (p_{loo}) was consistent with the hierarchical structure of the models. Monte Carlo standard errors of the expected log predictive density were negligible, confirming stable posterior inference. These results indicate that the reported effects are robust and generalize beyond the observed data.

S13. Justification of the Modeling Strategy

The modeling strategy in Experiment 3 was designed to align statistical analysis with the conceptual claims of the paper. Rather than treating attraction as a unitary shift in target choice, we decomposed choice behavior into decoy elimination and target–competitor preference formation. This decomposition reflects the theoretical distinction between rejecting dominated options and forming relative preferences among non-dominated alternatives.

Hierarchical Bayesian logistic regression provides a natural framework for this analysis by accommodating trial-level predictors, individual differences, and uncertainty in parameter estimates. The use of weakly informative priors regularizes estimates without imposing strong theoretical assumptions, while ROPE-based inference allows direct assessment of whether observed effects are practically meaningful rather than merely statistically detectable.

Model comparisons and robustness checks were included to demonstrate that the substantive conclusions do not depend on specific modeling choices. Taken together, this approach ensures that claims about asymmetric comparative processing are supported by transparent, conservative, and theoretically aligned statistical analyses.

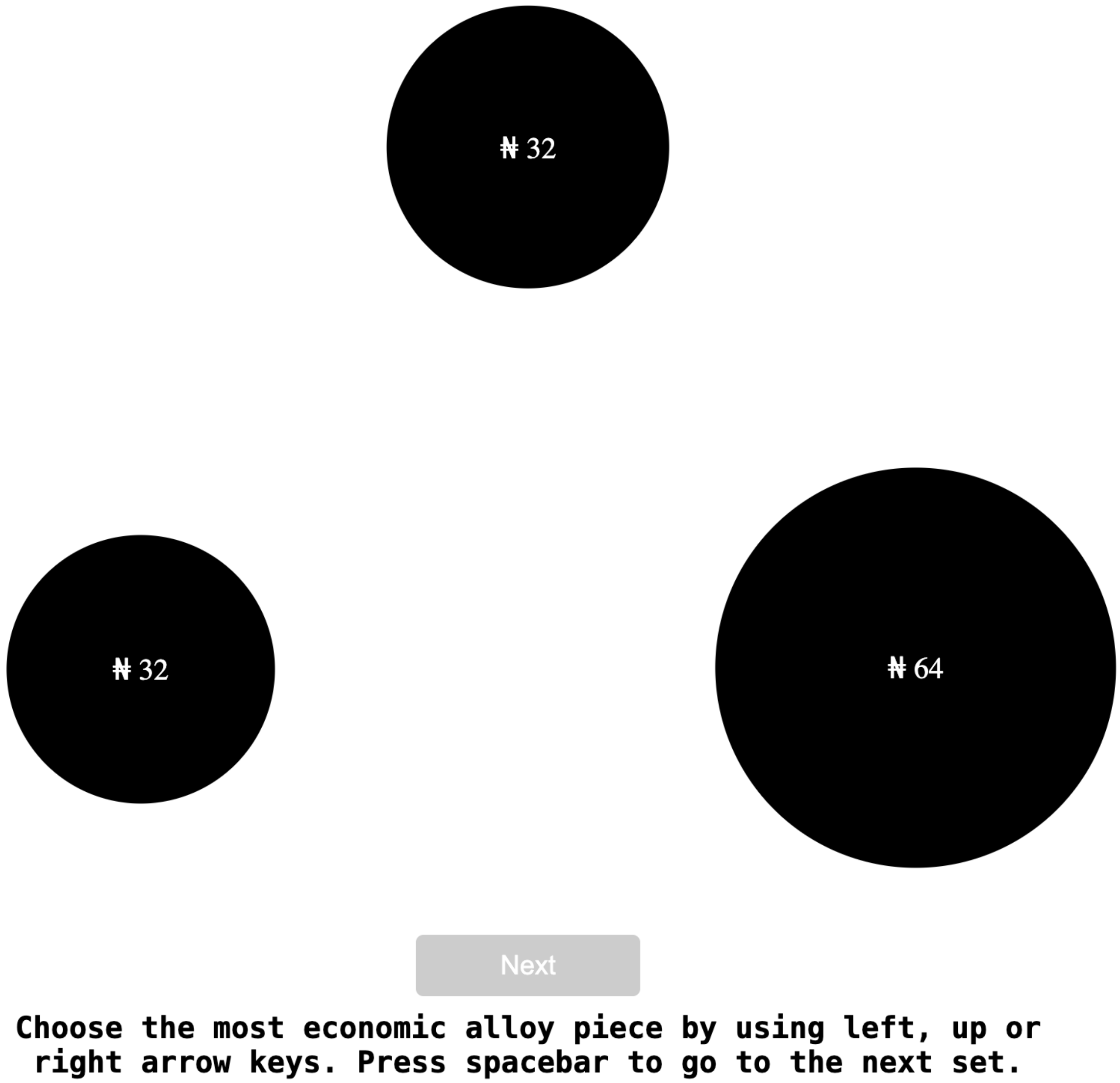

Figure 1

Example Trial in Experiment 1

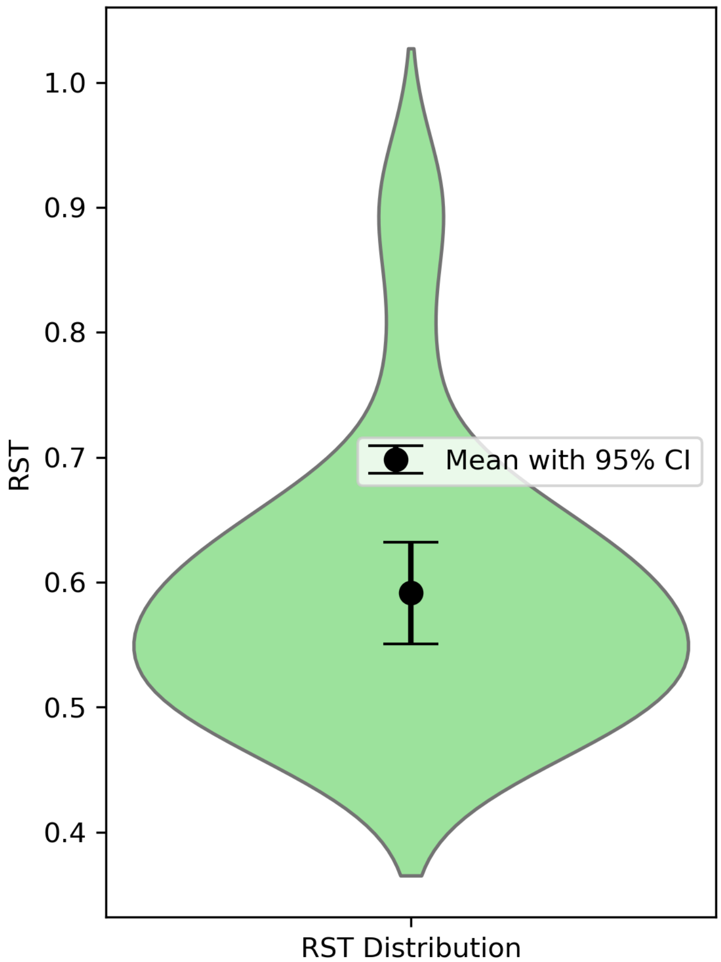

Figure 2

RST Distribution in Experiment 1

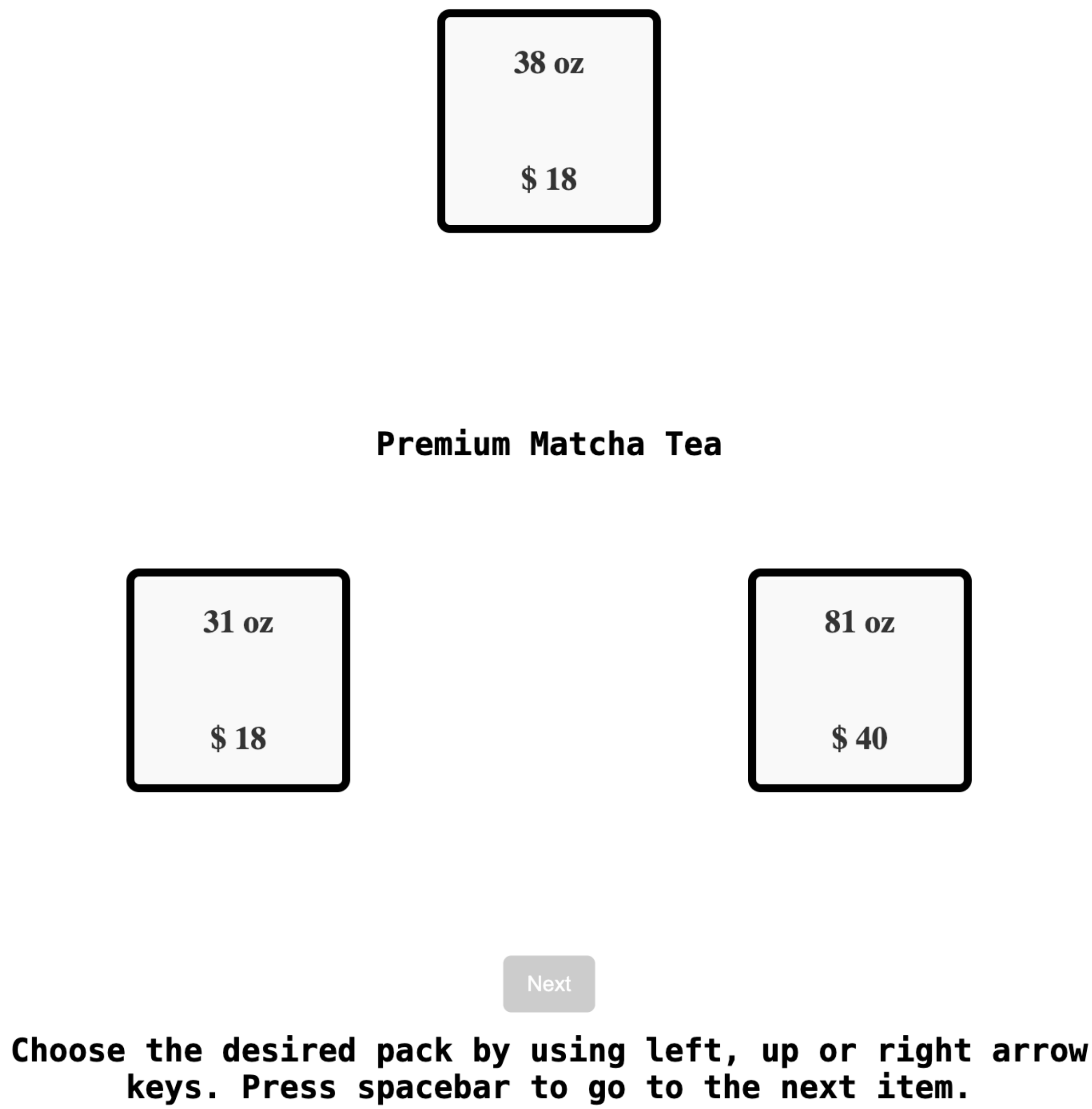

Figure 3

Example Trial for Block H in Experiment 3

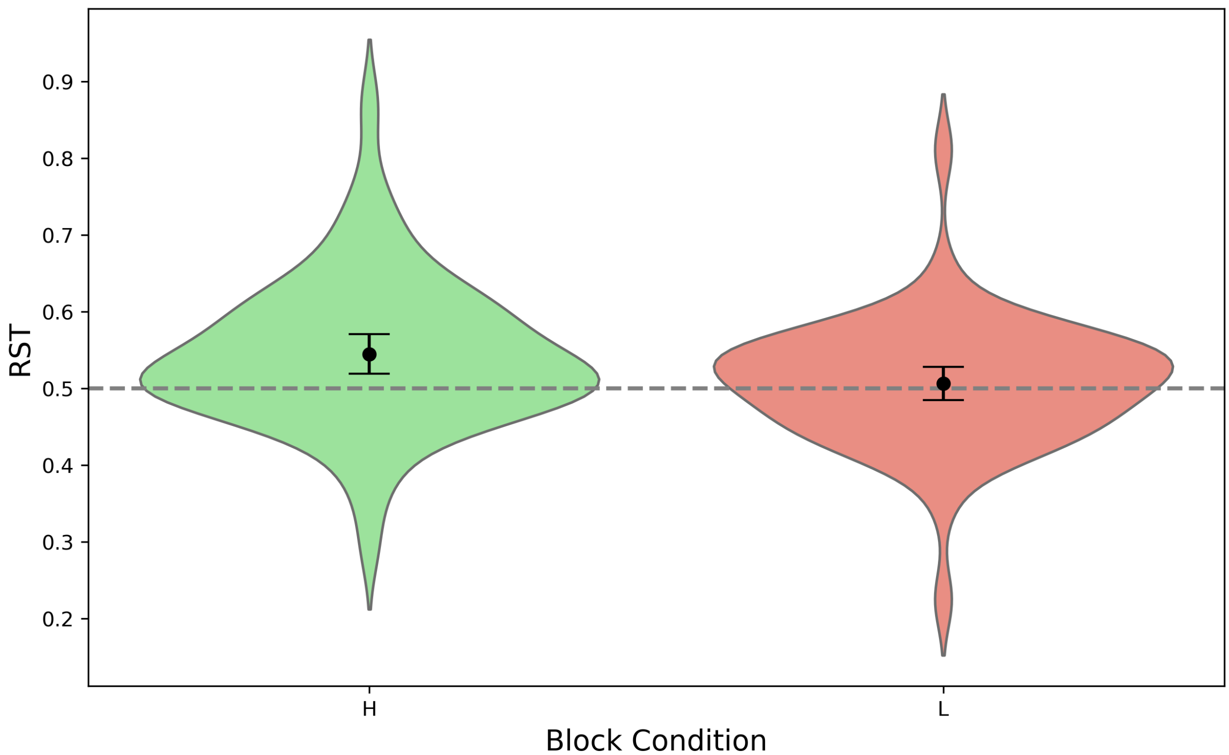

Figure 4

RST Distributions for two blocks in Experiment 3

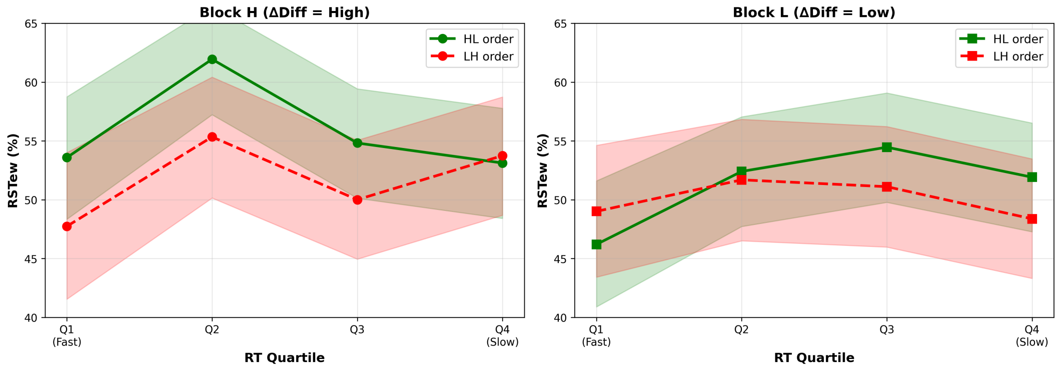

Figure 5

Relative share of the target (RST) across response-time quartiles, plotted separately by ΔDiff (high vs. low) and block order. Error bars indicate 95% credible intervals.

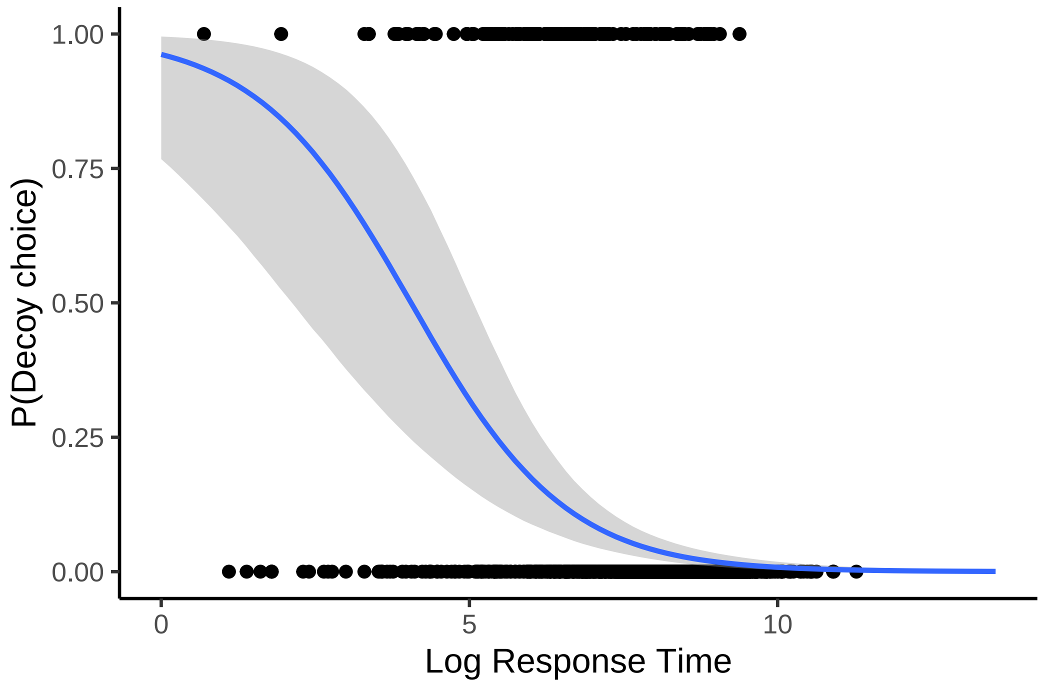

Figure 6

Model-predicted probability of decoy choice as a function of log response time. Shaded regions indicate 95% credible intervals.

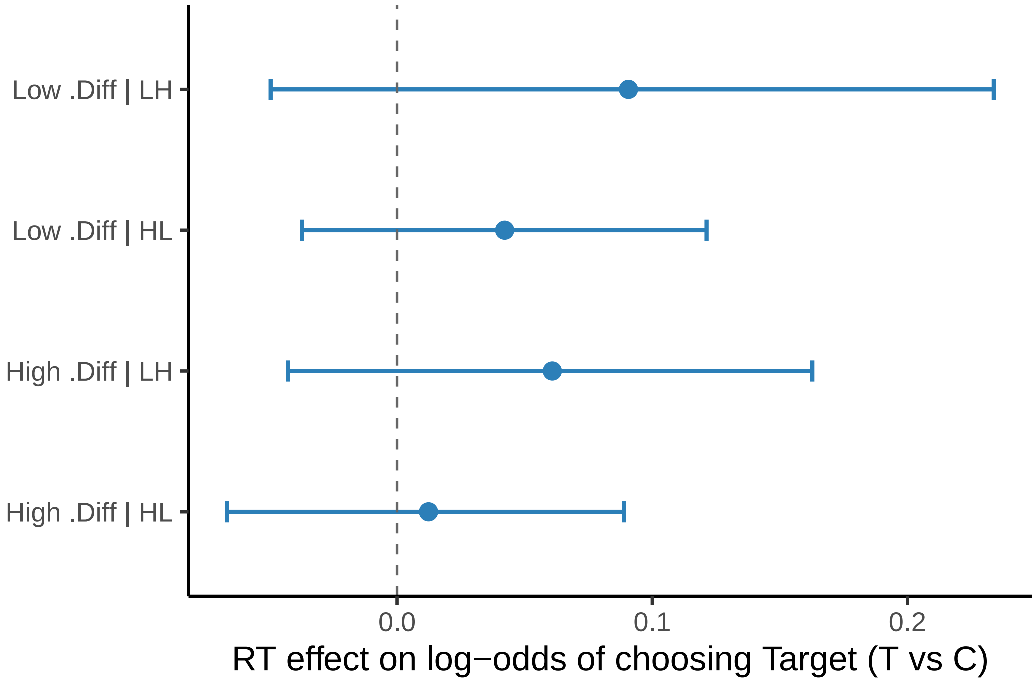

Figure 7

Posterior estimates of the RT slope on target choice (Target vs. Competitor) across △Diff and block-order conditions. Points denote posterior means; error bars indicate 95% credible intervals. The dashed line marks zero (no RT dependence).

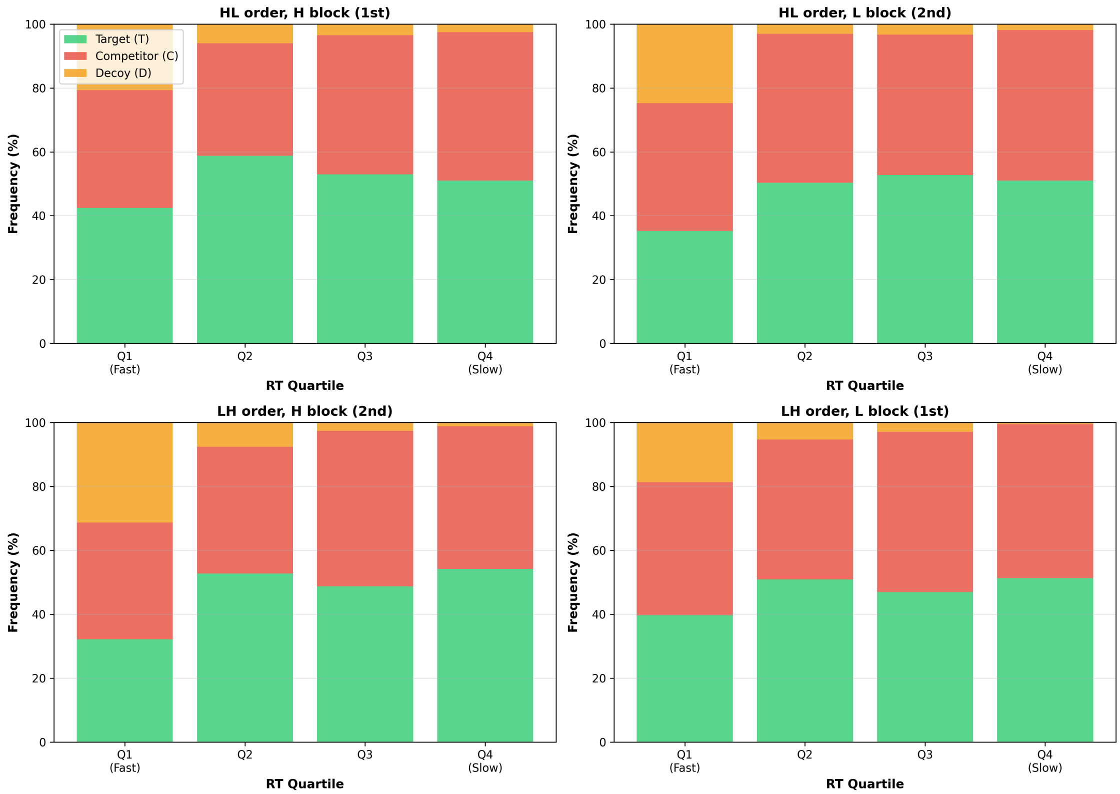

Figure 8

Choice proportions for Target (T), Competitor (C), and Decoy (D) across response-time quartiles. Decoy choice decreases monotonically with RT, whereas Target and Competitor choices increase symmetrically.

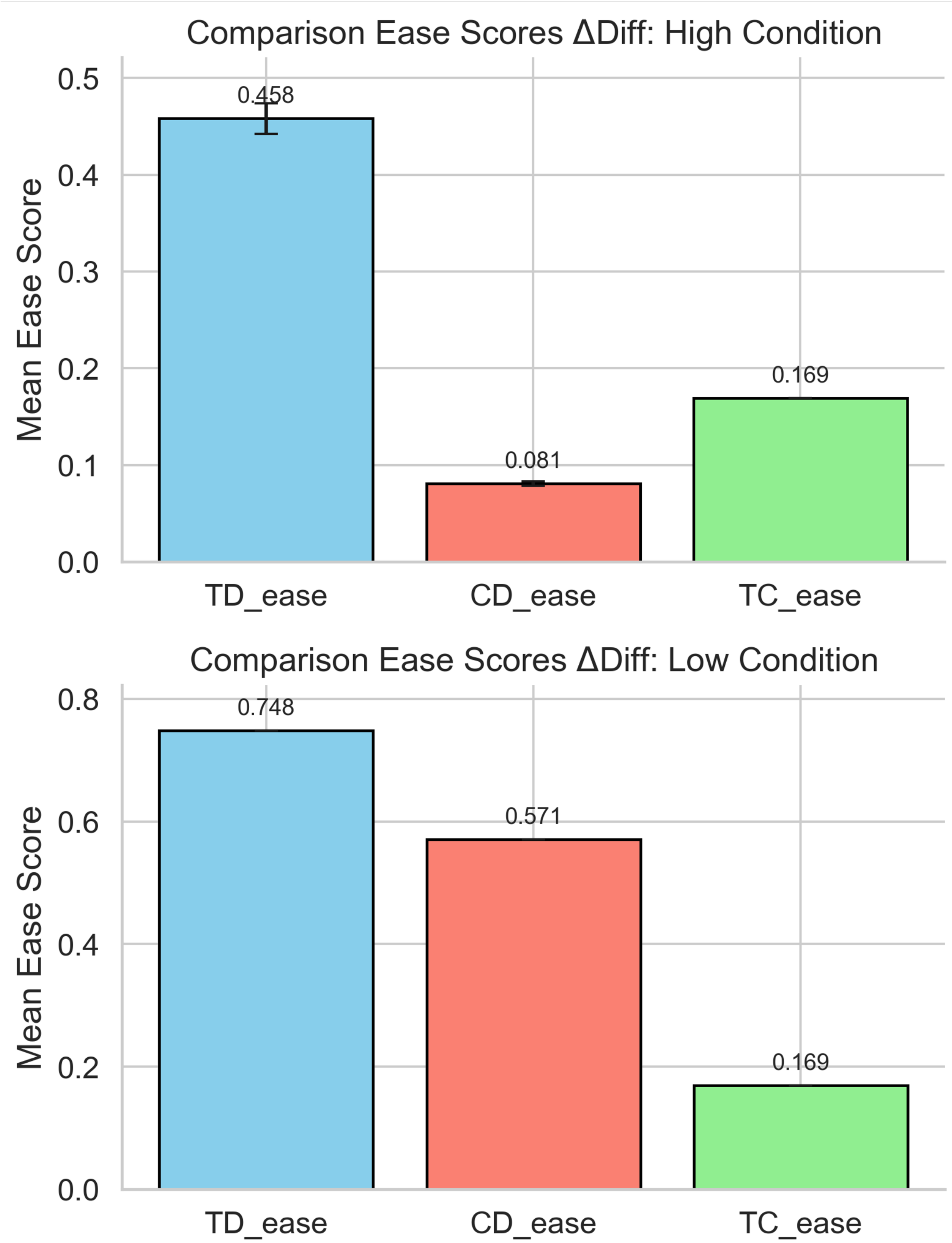

Figure 9

Figure S1. *Schematic illustration of pairwise comparative asymmetry manipulation. Panels show the distribution of ease-of-comparison scores for Target–Decoy (TD), Competitor–Decoy (CD), and Target–Competitor (TC) pairs across Low and High ΔDiff conditions. In the Low ΔDiff condition, TD and CD comparisons are matched in ease,*